

Rahul Deepak Kuchipudi

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Education

Indian Institute of Technology Bombay

2021 – 2025

Bachelor of Technology in Computer Science and Engineering

Achievements

All India Rank 157 in JEE Main amongst 1.2 million candidates

2021

All India Rank 709 in JEE Advanced amongst 1,50,000 candidates

2021

National Talent Search Scholarship awarded by NCERT, Govt. of India

since 2019

Experience

Software Development Engineer

Jul 2025 – Present

Avathon

Bengaluru, India

- **Leading the ground-up rewrite of IRIS**, Avathon's production video-analytics platform (deployed across 10+ edge sites), migrating from **DeepStream 6.2** → **9.0** — spanning C++/GStreamer pipeline, dynamic bundler, MQTT cloud integration, and use-case modules (door, svm-cv, hse-cv).
- Architected a **VSS (Video Summarization Service)** integrating **vLLM**-served VLMs and LLMs via Kafka streaming, with configurable caption-summarisation and aggregation prompts for domain-specific real-time scene understanding.
- Diagnosed and fixed cross-version memory leaks across **DS6.2** and **DS8** (GLib-managed vs. raw-malloc classifier_type regression; opTensorFilePath leak); decoupled TensorRT engine batch-size from dynamic camera count, eliminating full engine rebuilds on fleet changes.
- Improved **inference throughput by 40%** on RTX-class edge GPUs via TensorRT FP16/INT8 calibration, batched preprocessing, and zero-copy buffer management through the **NvBufSurface** API; authored custom C++ GStreamer/DeepStream plugins for GPU-native NV12→BGR conversion and per-frame metadata extraction.
- Built an **automated DS9 + Triton benchmark harness** (Docker Compose, pyservicemaker, pynvml) producing FPS/latency/GPU-memory reports across YOLO model families on RTX 3060 edge hardware; hardened pipeline with exponential-backoff retry and boundary-safe OpenCV bounding-box handling.

Software Developer Intern

May 2024 – Jul 2024

Hilti Technology Solutions India

India

- **Automated** MSI installer creation via CI/CD pipeline using GitLab CI, Docker containers, and WiX Toolset CLI, **reducing deployment time from 2 hours to 10 minutes** and eliminating manual errors across release cycles.
- Developed and implemented **automated test scripts** for comprehensive web application testing using **Selenium WebDriver** framework in C# with Page Object Model design pattern, improving test coverage and regression detection.

Projects

Video Scene Summarisation with VLM + LLM | *Python, Kafka, PyTorch*

Jan 2026 – Present

- Designed an end-to-end **video summarisation pipeline** combining Vision-Language Models for frame-level scene understanding and Large Language Models for generating structured natural language summaries.
- Implemented **Kafka-based streaming architecture** with dedicated topics for frame extraction, VLM inference, and LLM summarisation stages, enabling modular, fault-tolerant, and horizontally scalable processing.
- Built configurable **prompt templates** for domain-specific summarisation (security, retail, traffic monitoring) with structured JSON output consumed by downstream analytics APIs.
- Deployed on multi-GPU servers with **Docker**, supporting distributed inference across edge and cloud nodes.

Real-Time Multi-Stream Video Analytics Engine | *C++, Python, DeepStream*

Aug 2025 – Present

- Engineered a **GPU-accelerated video analytics system** processing 8+ concurrent live video streams at 30+ FPS using NVIDIA DeepStream SDK with custom GStreamer plugins and intelligent frame sampling algorithms.
- Built custom **C++ preprocessing plugins** with NvBufSurface API for batched inference, zero-copy GPU memory operations, and RTSP-based video chunking for **24/7 edge deployment** with automatic recovery mechanisms.
- Optimised pipeline stability, memory usage, and GPU utilisation for continuous inference workloads with comprehensive logging and production monitoring dashboards.

Semi-Supervised Skin Lesion Classification | *PyTorch, ResNet-50*

Aug 2023 – Nov 2023

- Built a **semi-supervised deep learning pipeline** using MixMatch with ResNet-50 backbone for 7-class skin cancer classification from 10,000 dermatoscopic images. *Guide: Prof. Preethi Jyothi, IIT Bombay.*
- Applied **pseudo-labelling**, consistency regularisation, and mixup augmentation to effectively leverage 8,000+ unlabelled images alongside limited labelled data, with custom PyTorch data loaders and augmentation pipelines.

Technical Skills

Languages	Python, C++, C, Bash, SQL
AI/ML	PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, OpenCV, Hugging Face
Inference & Edge	NVIDIA DeepStream SDK, TensorRT, CUDA, GStreamer, ONNX Runtime
DevOps & Tools	Docker, Linux, Git, GitLab CI/CD, GPU Programming